CANALRUNG · USER REQUIREMENTS

CanalRung user requirements

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	CNR-URS-001	Version	0.1
Status	Draft	Date	2026-11-05
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/canalrung		
Source	docs/03a-user-requirements.md @ ce4cb4e		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-05	Amish Chadha	First user requirements (kit 1.9.1 upgrade), written from the users and stakeholders of CNR-PRB-001 Table 1

CanalRung user requirements

What the people in CNR-PRB-001 need, written from their side. The product requirements that meet these needs are in CNR-PRS-001 (`03-requirements.md`) and the links are in the traceability matrix (`03b-traceability.md`). Method: STANDARDS section 22. No user has been interviewed yet: the needs come from the problem statement and the sources it cites, and a co-design partner (CNR-PRB-001, Co-design) should confirm them.

> **Safety:** CanalRung is a research and educational design for rescue equipment, never certified life-saving equipment. Never enter the water to rescue someone: throw, anchor and call the emergency number.

Table 1. User requirements.

ID	Category	Title	Need	Priority	Source
UR-001	Performance	Reach the person	A bystander can put something within the person's reach in a lined canal where the water is up to 2.0 m below the coping on a 1.5:1 slope, or 3.0 m on a vertical wall	High	Person in the water; Bystander on the bank (PRB Table 1)
UR-002	Deployment and assembly	Quick and obvious to use	An untrained bystander gets it into the water in well under a minute, guided only by what is printed on the box	High	Bystander on the bank (PRB Table 1)
UR-003	Usability	Climb out unaided	A person who is cold and tired can use it to climb a steep, slick wall without help	High	Person in the water (PRB Table 1)
UR-004	Safety	Carries an adult	It holds a heavy adult climbing out, with a margin	High	Person in the water (PRB Table 1)
UR-005	Safety	Stays attached to the bank	It does not pull free when a person holds on in a current	High	Person in the water; Bystander on the bank (PRB Table 1)
UR-006	Safety	Harms no one	Nothing that is thrown or deployed can hurt the person in the water or the helper	High	Person in the water; Bystander on the bank (PRB Table 1)
UR-007	Usability	One adult can handle it	A single adult, not necessarily strong, can lift and throw or place it	Medium	Bystander on the bank (PRB Table 1)
UR-008	Performance	Within reach at the surface	The part the person grabs stays at the water surface and does not sink	Medium	Person in the water (PRB Table 1)
UR-009	Environment and durability	Lasts outdoors	It stays ready through at least two years of sun, rain and frost without upkeep	Medium	Irrigation district and farm managers (PRB Table 1)
UR-010	Cost	Cheap enough to place often	A district can afford a station at every crossing, pump house and path	Medium	Irrigation district and farm managers; Village committees and schools (PRB Table 1)
UR-011	Maintenance and logistics	Built and mended locally	A village workshop can build and repair it from common stock and hand tools	Medium	Village committees and schools (PRB Table 1)
UR-012	Maintenance and logistics	Inspected at a glance	An inspector can see in seconds that a station is complete and untouched	Low	Irrigation district and farm managers (PRB Table 1)
UR-013	Regulatory	Honest about what it is	Anyone who finds it knows it is an open research prototype that is not certified life-saving equipment	Medium	Irrigation district and farm managers (PRB Table 1)

Categories: deployment and assembly, performance, usability, safety, environment and durability, logistics, maintenance, cost, regulatory. Priorities: High, Medium, Low.